

AMENDMENT NO. 1 NOVEMBER 2025
TO
IS 4454 (Part 1) : 2001 STEEL WIRE FOR MECHANICAL SPRINGS —
SPECIFICATION
PART 1 COLD DRAWN UNALLOYED STEEL WIRE
(Third Revision)

(Page 1, clause 1) — Substitute the following for the existing:

‘This standard (Part 1) covers the requirements of cold drawn unalloyed steel wires, both round and shaped, for the manufacture of mechanical springs for static duty and dynamic duty applications. Patenting, prior to drawing, may be carried out as and when required.’

(Page 1, clause 2) — Substitute the following for the existing references:

<i>IS No.</i>	<i>Title</i>
IS 228 (various parts)	Methods for chemical analysis of steels
IS 1608 (Part 1) : 2022/ ISO 6892-1 : 2019	Metallic materials - Tensile testing — Part 1 : Method of test at room temperature (<i>fifth revision</i>)
IS 1717 : 2018 /ISO 7800 : 2012	Metallic materials — Wire - Simple torsion test (<i>fourth revision</i>)
IS 1755 : 2018 /ISO 7802 : 2013	Metallic Materials — Wire — Wrapping test (<i>second revision</i>)
IS 1956 (Part 3) : 2019	Glossary of terms relating to iron and steel : Part 3 Long products (Including Bars, Rods, Sections and Wires) (<i>second revision</i>)
IS 4163 : 2021/ISO 4967 : 2013	Steel — Determination of content of non-metallic inclusions — Micrographic method using standard diagrams (<i>fourth revision</i>)
IS 4905 : 2015 /ISO 24153 : 2009	Random sampling and randomization procedures (<i>first revision</i>)
IS 6396 : 2023 /ISO 3887 : 2017	Steels — Determination of the depth of decarburization (<i>third revision</i>)
IS 8910 : 2022 / ISO 404 : 2013	Steel and steel products — General technical delivery requirements’

(Page 1, clause 3.1) — Substitute ‘IS 1956 (Part 3)’ for ‘IS 1956 (Part 5)’.

(Page 1) — Insert the following new clauses after clause 3.6.

‘3.7 Shaped Wire

Wires having non-circular cross-section. Some of the common types of shaped wire are given below:

- a) Flattened wire — A wire having a cross section with two parallel sides (*see* Fig. 1a) which are much longer than the other rounded sides;
- b) Rectangular section wire — A special case of flattened wire, having a cross section with two sets of parallel sides at right angles to each other, but with one set longer than the other (*see* Fig. 1b);
- c) Square section wire — A wire having a cross section with two sets of parallel sides of equal length at right angles to each other (*see* Fig. 1c);
- d) Trapezoidal section wire — A wire having a cross section with two parallel sides, one shorter than the other, and with the other two sides of equal length. The special feature of this wire is that, when coiled into a spring such that the shorter minor characteristic dimension is located on the inside of the coil, the cross section of the material takes on either an approximately square or rectangular cross section. (*see* Fig. 1d); and
- e) Oval (elliptical) section wire — Wire with a cross round cross section (without edges), symmetrical about the center, with one minor axis (Shorter diameter) and one major axis (Longer diameter) (*see* Fig. 1e).

3.8 Equivalent Round Diameter

Diameter of a round wire having equivalent cross-sectional area to a given shaped wire.

NOTE — For shaped wires, cross sectional area shall be calculated by measuring width and thickness and multiplying by a shape factor.

3.9 Shape Factor

A value used to obtain cross sectional area for shaped wires when multiplied by measured width and measured thickness.

NOTE — The shape factor value shall be agreed upon between the supplier and the purchaser.

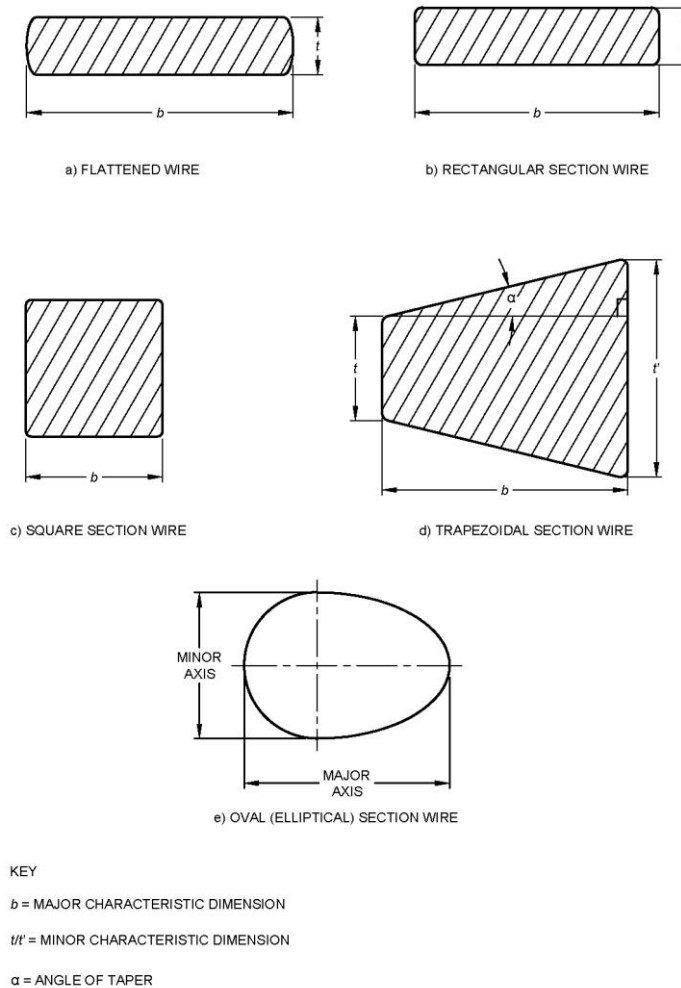


FIG 1. CROSS SECTIONS OF TYPICAL NON-CIRCULAR WIRE SHAPES INDICATING CHARACTERISTIC DIMENSIONS'

(Page 3, clause 9.3) — Substitute the following for the existing clause:

'9.3 Tolerances for size other than those specified in Table 4 may be mutually agreed upon between the purchaser and the manufacturer.'

(Page 3, clause 9) — Insert the following clause and new table after clause 9.3:

'9.4 Unless otherwise agreed upon, the size and tolerance on characteristic dimensions of the shaped wires shall be in accordance with Table 5.

Table 5 Tolerance on Characteristic Dimensions of Shaped Wires

(Clause 9.4)

Sl No.	Nominal Dimension, mm		Square, Rectangular and Trapezoidal Sections, mm	Flattened Wire	
	Over	Upto and Including		Tolerance on Width, mm	Tolerance on Thickness, mm
(1)	(2)	(3)	(4)	(5)	(6)
i)	—	1.0	± 0.03	± 0.07	± 0.020
ii)	1.0	2.0	± 0.04	± 0.08	± 0.025
iii)	2.0	3.0	± 0.05	± 0.09	± 0.030
iv)	3.0	4.0	± 0.06	± 0.10	± 0.040
v)	4.0	5.0	± 0.07	± 0.11	± 0.050
vi)	5.0	6.0	± 0.08	± 0.12	± 0.060
vii)	6.0	8.0	± 0.09	± 0.13	± 0.070
viii)	8.0	10.0	± 0.10	± 0.14	± 0.090
ix)	10.0	12.0	± 0.11	± 0.15	± 0.090
x)	12.0	—	± 0.12	± 0.16	± 0.100'

(Page 3, clause 10.1) — Substitute the following for the existing:

‘10.1 Tensile Test

The tensile test shall be carried out in accordance with IS 1608 (Part 1). The tensile strength and reduction in area of the wire shall be as given in Table 6 and shall be calculated using the actual wire diameter for round wires and using equivalent round diameter (*see 3.8 and 3.9*) for shaped wires. Reduction in area, percent, of shaped wires may be mutually agreed upon between the purchaser and the supplier.

NOTE — Measure the maximum and minimum dimension on the necked down section and multiply by the shape factor to estimate the cross-sectional area for use in the standard reduction of area calculation.’

(Page 3, clause 10.2) — Substitute the following for the existing:

‘10.2 Wrapping Test

Wrapping test shall be applied to wires with nominal diameter less than 3.00 mm. Wrapping test shall be carried out in accordance with IS 1755. The wire shall not show, on visual examination, any sign of fracture when closely coiled for at least four turns around a mandrel of diameter equal to that of the wire. Wrapping test is not applicable to shaped wires.’

(Page 4, clause 10.3.1) — Insert the following at the end:

‘If specified, torsion test requirements for shaped wires may be agreed upon between the purchaser and the supplier’

(Page 4, clause 10.3.3, line 3) — Substitute ‘Table 7’ for ‘Table 6’.

(Page 4, Table 5, title) — Substitute ‘Table 6’ for ‘Table 5’.

(Page 6, Table 6, title) — Substitute ‘Table 7’ for ‘Table 6’.

(Page 6, clause 10.4) — Insert the following at the end:

‘For shaped wires, bend test shall be over the major dimension, unless otherwise agreed between the purchaser and supplier at the time of order. Wire shall be bent through 90° around a mandrel of following diameter:

- a) Twice the major characteristic dimension, if the same is between 3 mm to 6 mm; and
- b) Thrice the major characteristic dimension if the same is > 6 mm.

To avoid need for excessive number of mandrel sizes, wire shall be deemed to have conformed to this standard if it can be bent around a mandrel of diameter smaller than that specified. In carrying out this test, wire shall be free to move longitudinally in the forming device.’

(Page 6, clause 10.5) — Insert the following at the end:

‘For shaped wires, coiling test and requirements thereof, may be mutually agreed between the purchaser and the manufacturer.’

(Page 7, clause 10.6) — Insert the following new clause after clause 10.6.1:

‘10.6.2 For shaped wires, the requirements related to cast of the wire may be mutually agreed between the purchaser and the manufacturer.’

(Page 7, clause 10.6) — Substitute ‘Fig. 2’ for ‘Fig. 1’

(Page 7, Fig. 1) — Substitute ‘Fig. 2’ for ‘Fig. 1’ for the name of the figure.

(Page 7, clause 10.7) — Insert the following new clause after clause 10.7.1.1:

‘10.7.1.2 For shaped wires, deep etch test and requirements thereof, may be evaluated based on conversion to equivalent round wire diameter.’

(Page 7, clause 10.8.1) — Insert the following at the end:

‘For shaped wires, the partial decarburization is specified with the same % value as of round wires. The equivalent round diameter is used to calculate the maximum permissible depth of decarburization.’

(Page 8, clause 15.1.1) — Substitute the following for the existing:

‘15.1.1 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder and the products may be marked with the Standard Mark.’

(Page 8, clause 15.1.2) — Delete.

(Page 9, Annex C, clause C-2, line 2) — Substitute ‘Table 8’ for ‘Table 7’.

(Page 9, Annex C, clause C-3.1, Para 1, line 1) — Substitute ‘Table 8’ for ‘Table 7’.

(Page 9, Annex C, clause C-3.1, Para 2, line 4) — Substitute ‘Table 8’ for ‘Table 7’.

(Page 9, Annex C, clause C-3.2, Para 2, line 4) — Substitute ‘Table 8’ for ‘Table 7’.

(Page 9, Annex C, Table 7, title) — Substitute ‘Table 8’ for ‘Table 7’.